Commonly Used Silicones in our daily life

Silicone based sunscreens

OCEDERMA™ SILICONE SUNSCREEN GEL WATER RESISTANT AND NON- GREASY SUNSCRREN GEL SPF 50 PA+++ OIL FREE SUN PROTECTION AGAINST UVA/UVB RAYS ,

Silicone oil - Antifoam

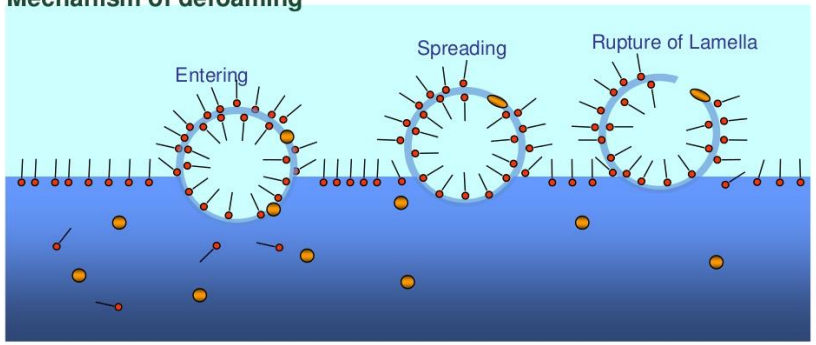
Defoamer – Basic definitions

150 years

BASF
We create chemistry

- A defoamer is a surface-active agent that destroys foam after it has formed. An antifoam agent is an additive that prevents foaming. The key to defoaming is to disrupt the surfactant stabilized foam lamella.

Mechanism of defoaming



4

BRANDS

TCC Materials®

Akona®

Spec Mix®

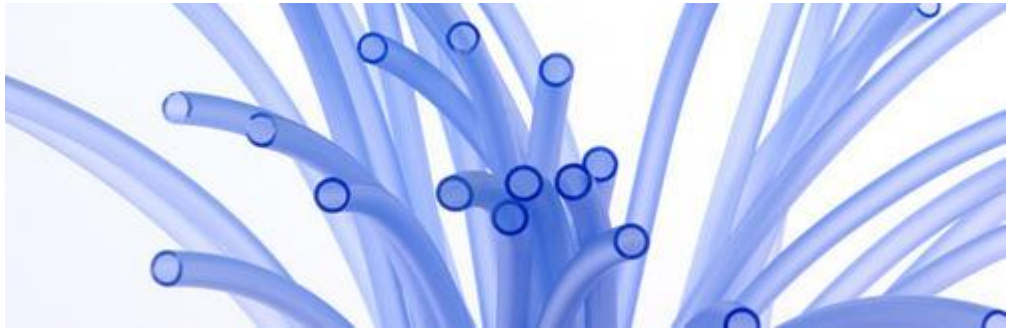
Quikrete®

TechMix®

ProSpec® Siloxane Sealer

ProSpec® Siloxane Sealer is a clear, solvent-based sealer for concrete and cementitious materials. It penetrates and reacts chemically within the capillaries to repel water and protect from the damaging effects of rain, water penetration, chemicals, salt, freeze-thaw cycles, and smog. Provides natural-looking, non-yellowing, invisible protection. Available in 1 gal. bottle (BOM #120826), 5 gal. pail (BOM #120825), and 55 gal. drum (BOM #120829).

Silicone - Elastomers

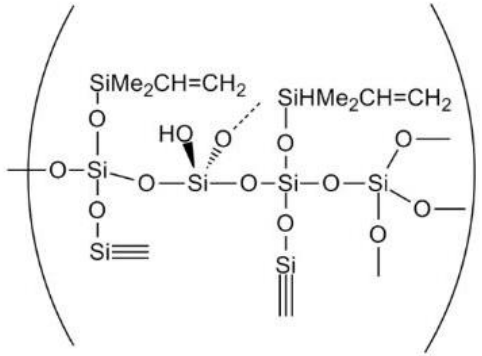


Silicone - Resin

Application: **Dental impression material**

Silicone: Ear Impression Material

XJY-8206 FOR DENTAL
METHYL VINYL MQ SILICONE RESIN VMQRESIN



Sealant (Low Molecular Weight Siloxane Reduction Type)

•Silicone Sealant Rtv (Room Temperature Vulcanising) Silicone Sealant

FEVISEAL NEUTRAL PRO one-part Neutral Cure Silicone Sealant is a non-slump, moisture-curing RTV (room temperature vulcanizing) that cures to form a tough, medium modulus rubber for long- term Fexibility and durability.

SPECIALTY GLUES

Some repairs require very specific custom-made adhesives.
Our Specialty adhesives are engineered for difficult to bond substrates like shoes, glass, plastics, nuts & bolts, vinyl and more.

Silicones are extensively used for food packaging like fruit labels, bakery papers, wrappers for candy, chewing gum, meat and frogen fruits.

Are fruit labels are safe to eat??

Yes, they are

Silicone Resins Market worth $6.3 billion by 2026 - Exclusive Report by MarketsandMarkets™

MARKETSANDMARKETS™

Material Properties of Silicone Rubber

Long Term Compression Set Resistance

Properly designed Silicone Foam and Silicone Sponge outdoor enclosure gaskets can effectively seal out wind driven rain and dust – helping the designer meet NEMA enclosure and IP specifications. Unlike most organic elastomers such as EPDM and neoprene, silicone maintains its resiliency over a broad temperature range and resists taking a permanent compression set.

Flame Retardant Capabilities

Silicone Rubber can be compounded to be flame retardant and achieve UL94V0, UL94V1 and UL94HF1 certifications. Flame retardant closed cell silicone sponge and silicone foam gaskets are used in analytical instrumentation, telecommunications equipment and controls used in Mass Transit systems. Even when silicone rubber burns under extreme temperatures, the by-products are non-toxic and any residual ash continues to provide electrical insulation properties.

FDA Regulations

Silicone Rubber can be compounded using ingredients that meet FDA regulations for properties such as high tear strength and flexing using platinum based catalysts. These silicone products are often specified for healthcare, instrumentation, food processing and medical diagnostic equipment applications.

Electrical Conductivity

Silicone Rubber can be blended with nickel-graphite or silver plated aluminum particles for electrically conductive gaskets that provide EMI (electro-magnetic interference) Shielding for electronic communication equipment. Conductive silicone is available in sheets or custom molded gaskets.

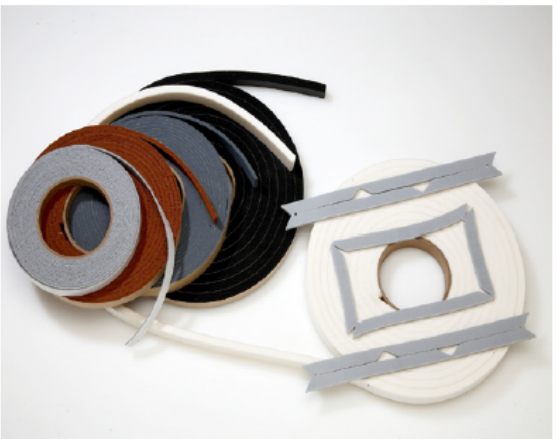
Electro-Static Discharge (ESD)

Solid silicone rubber can be blended with semi-conductive carbon for ESD properties. Further, closed cell silicone sponge will accept a surface coating of conductive silicone for soft gaskets and pads with ESD grounding properties.

Thermal Conductivity

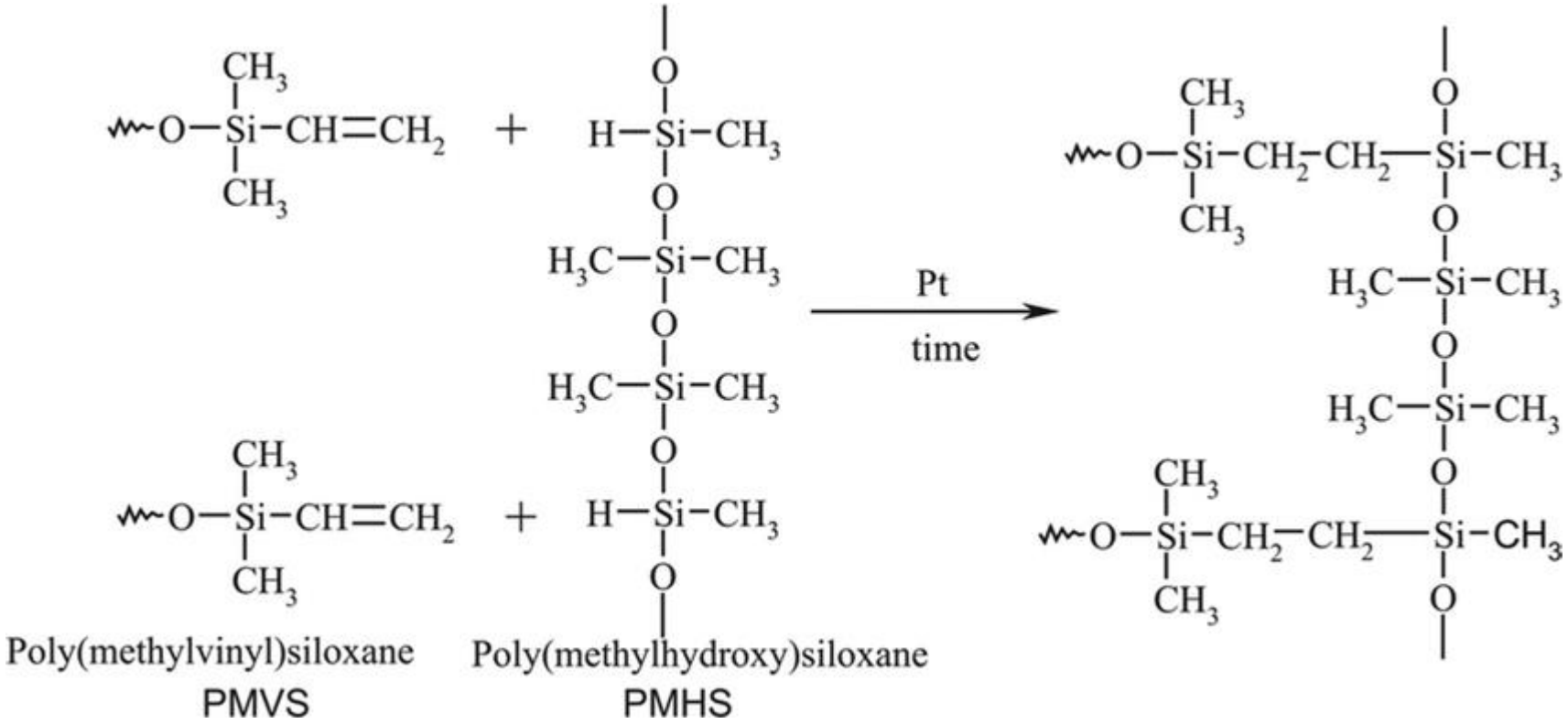
Silicone Rubber can be blended with ceramic powders such as alumina and boron nitride for thermal interface pads that help to dissipate heat from power generating devices in electronics. Thermal interface pads are typically fabricated from gel based silicones which are very soft and conformable.

Silicone Sponge Rubber and Silicone Foam Materials

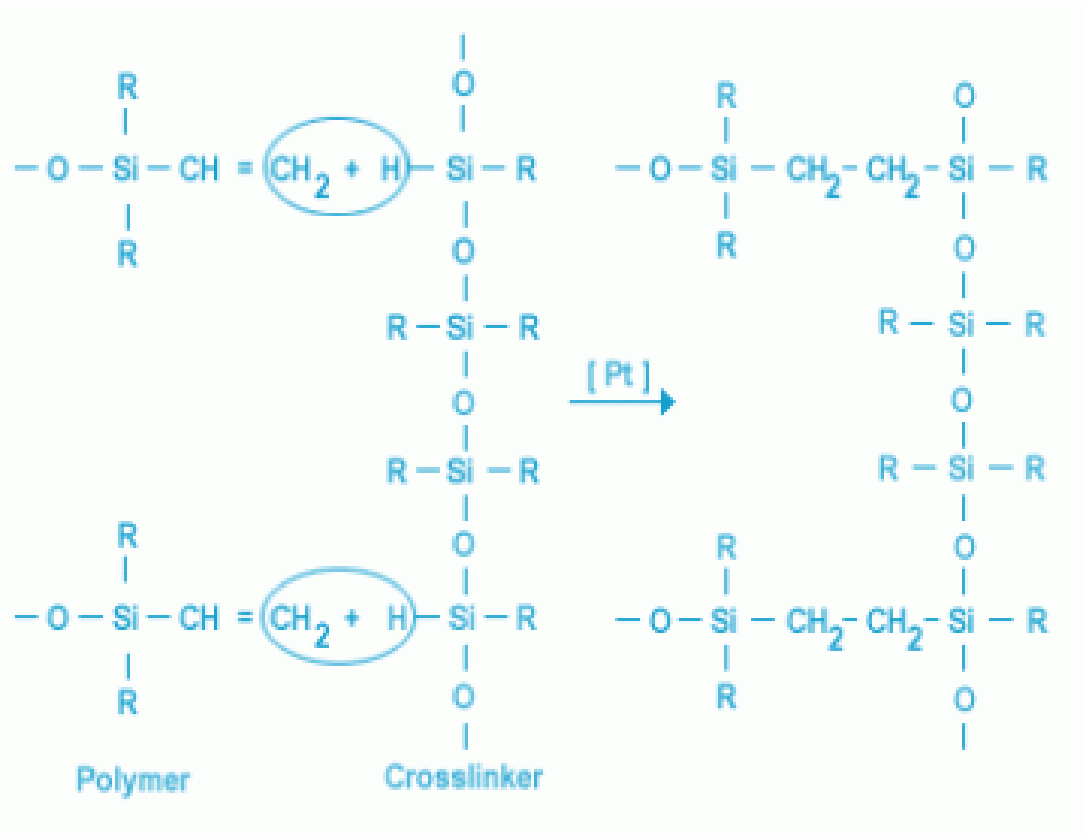


Closed Cell Silicone Sponge is heat cured from gum base silicone into molded sheets or rotocured into continuous rolls. High Quality silicone sponge is typically post-cured to improve mechanical properties and high temperature performance. Silicone sponge has a characteristic fabric impression on the skin surfaces. Silicone Sponge generally has better tear and tensile strength than cast silicone foams, making molded sponge a better choice for demanding mechanical applications.

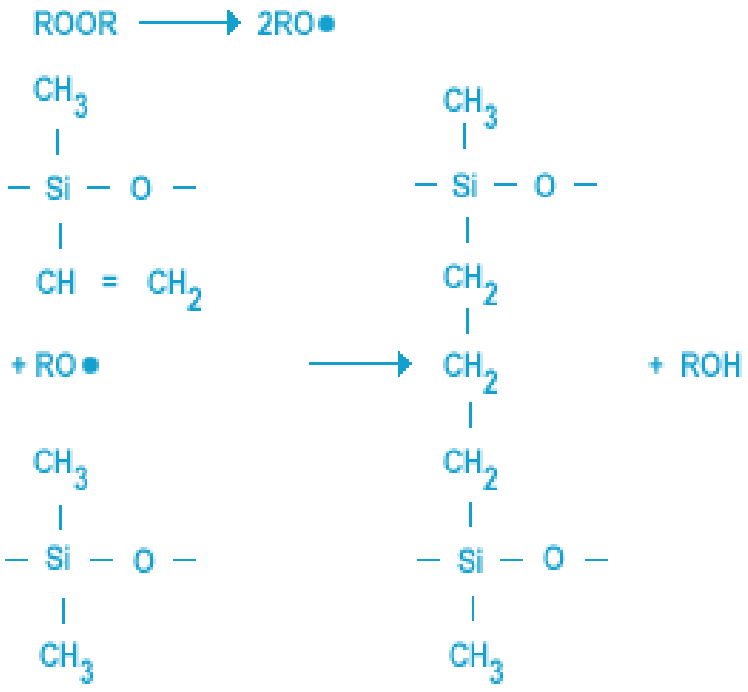
Silicone Foam sheet materials are cast from liquid silicone compounds - which are catalyzed, rapidly mixed, cast onto a release liner and then cured at elevated temperatures. Many silicone foam materials pass UL94 flame ratings. Silicone foams have a smooth surface on one or both surfaces and can be provided from .032″ through .500″ thick in continuous rolls. Soft, low density grades of silicone foam are open cell foams, however the medium and higher density silicone foams can function as effective gaskets for sealing out weather and wind driven rain when enclosures are designed for appropriate gasket deflection.



Platinum Curing of a Silicone Polymer



Platinum Curing of a Silicone Polymer



Peroxide Curing of a Silicone Polymer

https://www.tblplastics.com/platinum-vs-peroxide-silicone-tubing/